Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Early-career technology professional with 2+ years delivering **software, data, AI, cloud, and automation solutions**; combines analytical problem-solving, continuous learning, cross-functional communication, and ownership to build scalable, secure, maintainable systems with measurable business value.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Designed **cloud data and software solutions** across FastAPI, PostgreSQL, SQL Server, and Azure storage, supporting 500+ users while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Evaluated alternative workflow designs and implemented fault-tolerant AI automation for 1,000+ jobs, improving throughput 3x while delivering 5,000+ monthly integrations with validation and 99%+ production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built **full-stack React and Python solutions** for uploads, processing, history, and reporting, serving 150+ users and 10,000+ daily requests while reducing daily feedback turnaround time 60%.
- Integrated OpenAI, Claude, and Qwen through adaptable APIs, researching scalable alternatives and implementing secure validation, cost tracking, fallback handling, and production CI/CD releases for reliable AI-enabled workflows.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **cloud, infrastructure, and data integrations** using AWS Glue, S3, SQL, APIs, and caching, improving response times 30%, increasing pipeline efficiency 68%, and reducing technical debt 20%.
- Collaborated with Microsoft and GitHub stakeholders to clarify requirements, assess implementation options, review code, and deliver reliable cross-platform automation with 20+ tests across Windows, Linux, and macOS.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **product and business technology** across eight production React and Node.js services for authentication, records, billing, permissions, notifications, and reporting, securely synchronizing 2M+ records through Azure SQL.
- Improved service operations using automated testing, Sentry, Azure Monitor, Kibana, Docker, and Jenkins, achieving 95%+ overall build validation while reducing deployment time 50% and resource wastage 42%.

Cyber Security and Data Mining **India**
Data Analyst - Research Fellowship *May 2023 – August 2023*

- Designed a **comprehensive large-scale cybersecurity analytics pipeline** over 650,000+ URLs, extracting 21 behavioral indicators and comparing statistical, machine-learning, and deep-learning approaches to identify scalable detection alternatives.
- Achieved 96%+ malicious-URL detection accuracy using Gaussian Naive Bayes, Random Forest, RNN, LSTM, Bi-LSTM, GRU, and CNN models while communicating actionable phishing and malware findings clearly to stakeholders.

Skills

Programming: Python, Java, JavaScript, TypeScript, SQL, C++, Bash, React.js, Node.js, FastAPI
Cloud & Platform: AWS, Azure, GCP, Docker, Kubernetes, Linux, microservices, REST APIs, CI/CD
Data & AI: PostgreSQL, SQL Server, MongoDB, ETL, analytics, machine learning, LLMs, automation
Security & Operations: Authentication, access control, testing, monitoring, logging, troubleshooting, Sentry, Kibana
Architecture & Delivery: System design, requirements analysis, Git, Jenkins, GitHub Actions, Agile/Scrum, documentation

Projects

Big-Brain Live | *AI, PostgreSQL, GCP, Data Pipelines*

- Built a **secure AI and cloud data platform** processing 50,000+ documents with authenticated uploads, incremental indexing, PostgreSQL metadata, semantic retrieval, and collaborative workflows, reducing retrieval latency 40% and review time 60%.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Enhanced a **reliable enterprise-style database application** supporting Oracle and 10+ SQL and NoSQL systems with schema exploration, structured execution, AI-assisted reporting, Docker, Maven, and cross-platform development workflows.

FinNexus Live | *Next.js, Kubernetes, MongoDB*

- Developed a **scalable financial technology platform** supporting 50,000+ users and 1M+ transactions with secure integrations, real-time updates, MongoDB persistence, advanced filtering, and Kubernetes-based production CI/CD automation.